An LLM-Augmented Stacked Ensemble for Vietnamese Emotion Recognition on Social Media

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Abstract

Fine-grained emotion recognition from Vietnamese social-media text is a difficult natural language processing problem due to the language's low-resource status. This paper addresses a methodological gap on this problem: existing ensembles typically fuse base-model probabilities with fixed rules or linear stacking and do not condition on the raw sentence, while large language models are used mainly as offline generators or as static prompts. We propose a two-stage framework, evaluated on UIT-VSMEC, a seven-class Vietnamese social-media emotion corpus. In Stage 1, Gemini-2.5-Flash generates and paraphrases minority-class examples via 10-shot prompting. In Stage 2, five heterogeneous base classifiers (PhoBERT, CafeBERT, ViBERT, Logistic Regression, LinearSVC) produce probability distributions, and a QLoRA-fine-tuned meta-learning Gemma-2-9B-it consumes the raw sentence and those five distributions jointly. The full framework reaches 70.33% $\pm$ 1.33 accuracy, 70.39% $\pm$ 1.26 weighted F1, and 67.27% $\pm$ 1.25 macro F1. On the same settings, Logistic Regression stacking reaches 67.48% $\pm$ 0.50 weighted F1 and weighted voting reaches 67.91% $\pm$ 1.20.

***Index Terms*—Vietnamese NLP, emotion recognition, data augmentation, large language models, ensemble learning, social media**

I. Introduction

The rapid growth of Vietnamese social-media platforms has produced vast volumes of user-generated text carrying rich emotional signals. Mining these signals through fine-grained Emotion Recognition (ER) yields value for market research, opinion monitoring, and mental health screening. However, ER remains substantially harder than coarse sentiment polarity classification. Sentiment analysis on Vietnamese text, which is usually coarse-grained emotion (positive, negative, neutral), has benefited from larger annotated corpora, mature pre-trained language models, and relatively clear lexical polarity cues. Hybrid architectures such as PhoBERT combined with CNN-LSTM reach over 93% accuracy on the UIT-VSFC [1]–[3], and several deep-learning baselines report F1 scores above 90% on comparable polarity benchmarks. By contrast, fine-grained emotion recognition explores more aspects of the emotion, where labels are linguistically subtler and more context-dependent. On UIT-VSMEC, the best reported result from the original benchmark study reaches only 59.74% weighted F1 using CNN with word2vec embeddings [4]. Subsequent monolingual BERT models have narrowed but not closed this divide, indicating that scaling pre-training alone is insufficient for reliable fine-grained Vietnamese emotion classification.

Two obstacles drive this gap. First, Vietnamese is a low-resource language [5], [6]: annotated datasets are scarce. Second, UIT-VSMEC suffers from pronounced class imbalance: the dominant label *Enjoyment* constitutes roughly 28% of the training set, whereas minority emotions such as *Surprise* ($\sim$4.5%) and *Anger* ($\sim$6.9%) are severely underrepresented. Together, these constraints limit the robustness and generalization of individual classifiers.

There exists a methodological gap in which current ensemble methods typically combine class probabilities using fixed rules, such as voting, or through linear models like logistic-regression stacking. In contrast, large language models (LLMs) are predominantly employed as offline data generators or as static prompts. We address this gap with a two-stage framework. In Stage 1, Gemini-2.5-Flash [7] performs targeted data augmentation for minority emotion classes using 10-shot prompting. In Stage 2, five heterogeneous base classifiers produce probability estimates; a Gemma-2-9B-it [8] act as the meta learning model is then fine-tuned with parameter-efficient QLoRA [9] on the raw sentence and those distributions jointly.

The main contributions of this paper are:

1) An LLM-based augmentation protocol that generates and paraphrases Vietnamese social-media sentences for minority emotion classes with a controlled 10-shot prompting strategy.
2) A stacked ensemble in which a QLoRA-fine-tuned meta learning model Gemma consumes the raw sentence and five base-model probability distributions.

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II. Related Work

A. Vietnamese Emotion Recognition

Ho et al. [4] introduced UIT-VSMEC, a 6,927-sentence seven-class corpus, establishing a CNN+word2vec baseline of 59.74% weighted F1. Nguyen and Nguyen [10] reached 64.40% through multi-label regression with preprocessing and key-clause extraction. Pre-trained transformers further narrowed the gap: PhoBERT [11] underpins hybrid CNN-LSTM architectures that achieve 93.24% accuracy on the polarity task [3], but the fine-grained ER on informal social-media text remains harder. CafeBERT [12] (XLM-RoBERTa trained on Vietnamese Wikipedia and news) reports 66.12% F1 on VSMEC; ViSoBERT [13] reaches 68.37% weighted F1; and a GRU–CNN–BiLSTM–LSTM ensemble [14] attains 65.79% macro F1. Despite these advances, minority emotions such as Fear, Anger, and Surprise remain difficult to classify and leave room for data-centric and ensemble improvements.

B. LLM-based Augmentation and Ensemble Methods

Ubani et al. [15] show zero-shot ChatGPT generation outperforms EDA and back-translation on SST-2 and SNIPS; Whitehouse et al. [16] extend LLM augmentation to cross-lingual settings; Üveges and Ring [17] confirm 10-shot LLM prompting yields macro-F1 gains for underrepresented emotion classes. However, prior work treats the LLM as an offline generator only, or as a static zero-/few-shot classifier, and prompt design for fine-grained Vietnamese emotion categories remains underexplored. On the ensemble side, Thin et al. [18] obtain 94.03% weighted F1 on VSFC [1] via soft-voting over five Vietnamese transformers; Thin et al. [19] further evaluate zero- and few-shot LLM prompting for Vietnamese sentiment. This leaves a methodological gap for Vietnamese emotion recognition: using an LLM both to generate minority-class data and to combine heterogeneous ensemble probabilities while conditioning on the sentence. We address that gap with the two-stage framework in Section III.

III. Methodology

This study proposes an end-to-end pipeline for processing Vietnamese social media data and recognizing emotions in sentences, as illustrated in Fig. 1. It consists of four stages: LLM data augmentation, text preprocessing, base classifier training with cross-validation stacking, and meta-learning LLM model fine-tuning.

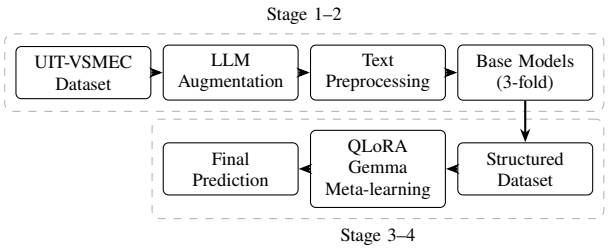


Fig. 1. Proposed pipeline: augmentation and preprocessing feed the base models; their cross validation probabilities form a structured dataset for QLoRA Gemma fine-tuning.

A. LLM-based Data Augmentation

UIT-VSMEC exhibits a severe imbalance between the most and least frequent classes. We address this by prompting Gemini-2.5-Flash [7] via LangChain [20] to add training examples for the underrepresented emotions, on the raw training split before any preprocessing. Two strategies are used (Table II): (i) *generated*—the model synthesizes new Vietnamese sentences from an emotion label and 10 randomly sampled in-context examples, creating scenarios absent from the original corpus; (ii) *paraphrased*—the model rewrites existing annotated sentences with linguistic variation while preserving the target emotion and original semantics. Table III shows examples of both. For each class below 1,500 examples, we generated enough candidates to reach a pre-deduplication target of 1,500; deduplication then left several final counts slightly below that threshold (Table I).

TABLE I
Class counts before augmentation and after deduplication.

	Enjoyment	Disgust	Other	Sadness	Anger	Fear	Surprise	Total
Before	1558	1071	1021	947	391	318	242	5548
After	1556	1498	1499	1500	1496	1493	1499	10541

Fig. 2 shows the six-stage pipeline. Batch generation uses temperature 0.9 [21] and a frequency penalty of 0.5.

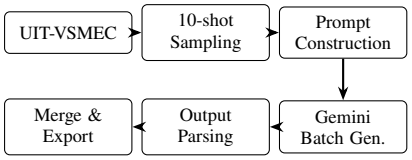


Fig. 2. Synthetic data creation pipeline for minority-class augmentation.

TABLE II
Prompt design comparison: generated vs. paraphrased strategy

	Generated	**Paraphrased**
System role	VN social-media expert; express target emotion only	Gen-Z social-media expert; preserve meaning, tone, intent
System task	Synthesize NEW sentences for the target emotion (in English and Vietnamese)	Rewrite sentences preserving SAME meaning and emotion
Key constraints	Natural VN style; length mix (5–10 / 10–20 / 20+ words); no other emotions	Semantic equivalence mandatory; no new facts or intensity; preserve negation and intensifiers
Output format	Numbered lines: sequence number, generated sentence, and example reference	

TABLE III
EXAMPLES OF AUGMENTED SENTENCES BY STRATEGY

Strategy	Original	Output	Emotion
Gen.	per không cho anh đi tập đâu nha mất bồ như chơi	Cái cảm giác bị bỏ rơi nó ám ảnh tui ghê lắm.	Fear
Gen.	sao mà mặt boss có thể gian phản diện đến thế nhỉ	Sao nhìn cái mặt của nhân vật phản diện nó gian manh vậy ta.	Surprise
Para.	sợ ngỗng còn hơn sợ chó :)))	Thà sợ chó còn hơn sợ ngỗng :)))	Fear
Para.	vl :)) còn đút tay túi áo ??	Trời đất ơi, còn đút tay vào túi áo nữa hả?	Surprise

B. Text Preprocessing

Vietnamese social-media text is noisy: users mix emojis, keyboard emoticons, slang, and short forms that carry strong emotional meaning. If these signals are removed too early, models may lose useful cues. Following practices similar to prior Vietnamese social-media emotion work [5], [10], we therefore apply a shared cleaning pipeline to the data after augmentation.

Each sentence is cleaned in four steps. First, a curated dictionary of emoji-to-Vietnamese mappings replaces Unicode emojis with short descriptive phrases (e.g., "vui vẻ" for happy faces and "buồn bã" for sad faces); unmatched emojis are then removed with a Unicode regex filter. Second, text emoticons such as `:))`, `:(`, and `T_T` are mapped to Vietnamese words, with longer patterns matched first to avoid partial replacements. Third, common internet abbreviations are expanded at the word level (e.g., *ko*/*k*/*hok*, all meaning *không*; *dc*/*đc*, meaning *được*). Fourth, URLs and HTML tags are stripped, non-Vietnamese characters are replaced with spaces while diacritics and basic punctuation are kept, repeated punctuation is collapsed, and whitespace is normalized.

After cleaning, duplicate sentences are removed so repeated content does not bias training. Stopwords are intentionally kept. Function words such as *không* (negation) and *rất* (intensifier) often change emotion meaning, and BERT-style encoders can use these tokens through self-attention [11], [12].

C. Base Classifiers

Five heterogeneous base classifiers—PhoBERT-v2 [11], CafeBERT [12], ViBERT [22], Logistic Regression [23], and LinearSVC—are trained with stratified 3-fold cross-validation, each producing probability vectors across the seven emotion classes.

Each model is also refitted on the full training set to produce validation and test probability arrays.

D. Stacked Ensemble with Meta-learning LLM

Weights and structured dataset. Each base model receives one weight, computed on the validation split after that model has been refit on the entire training set. For model i, Acc_i is the accuracy of the model on the validation data. The five accuracies are then scaled so the weights sum to one:

$$W_i = \frac{\mathrm{Acc}_i}{\sum_{j=1}^{M} \mathrm{Acc}_j}, \qquad M = 5. \tag{1}$$

A model that classifies more of the validation split correctly therefore takes a larger share of the weighted-average distribution. The five weights are calculated once, before LLM meta-learning training, and the same values are reused in the training, validation, and test prompts.

During meta-learning model training, each training sentence is paired with the probabilities distribution from the models, and W_i is applied to those out-of-fold distributions to form the weighted average. Validation and test prompts use the distributions from the full-training refit, because those sentences were held out of base-model fitting. Every prompt contains the original sentence, the five per-model distributions, and that weighted average, so the meta-learning model reads the text together with the ensemble signals.

QLoRA fine-tuning. Gemma-2-9B-it [8] serves as the predictor. The model is fine-tuned with QLoRA [9] using completion-only on the emotion label as the target output. At inference, the model generates the emotion label jointly conditioned on the sentence and the five base-model probability distributions in the structured prompt.

IV. EXPERIMENTS

A. Dataset and Splits

UIT-VSMEC [4] contains 6,927 human-annotated sentences drawn from Vietnamese social-media platforms, covering seven emotion labels: Enjoyment, Sadness, Fear, Anger, Disgust, Surprise, and Other.

B. Reproduction Settings

Table IV is the full setting list, and reported scores are the mean $\pm$ sample standard deviation over the three seeds.

TABLE IV
UIT-VSMEC HYPERPARAMETERS

Setting	Value
Seeds	42, 131, 9998; each seed is one full pipeline
Ensemble transformers	`vinai/phobert-base-v2`, `uitnlp/CafeBERT`, `FPTAI/vibert-base-cased`; 20 epochs, AdamW 2×10^{-5}, cosine, warmup 0.1, wd 0.01, clip 1.0; train/val batch: 128/256, accum. 2, max len 256; patience=3 on macro F1; 3-fold then full-train refit;
Standalone transformers	`uitnlp/CafeBERT`, `uitnlp/visobert` (`use_fast=False`); same optimizer, batch, and length;early stop, patience=5 on macro F1;
TF-IDF / linear	1–2 grams, min_df $=$ 2, sublinear_tf; LogReg and LinearSVC $C = 1.0$, random_state $=$ pipeline seed, 3 folds; LinearSVC + Platt scaling (inner $cv = 3$)
LogReg stacking	$C \in \{0.25, 1.0, 4.0\}$ by validation macro F1; L-BFGS, max_iter $= 1000$, seed 123
QLoRA Gemma	`google/gemma-2-9b-it`; pipeline seed; 4-bit NF4, double quant., bf16; $r = 16$, $\alpha = 32$, dropout 0.05, no bias; targets *q*/*k*/*v*/*o*/gate/up/down proj.; 2 epochs, lr 1×10^{-4}, cosine, 10% warmup, wd 0; batch 16, accum. 2, eval 16, seq 800, completion-only loss; greedy, max_new_tokens $=$ 12; LoRA inference truncates at 800; zero-shot max length 2048

C. Baselines and Evaluation

Evaluation uses accuracy, weighted F1, and macro F1. The augmentation trains CafeBERT on the original, paraphrased, and generated sets and reports Fear, Anger, and Surprise on the test set. The ensemble comparison is Table VIII. Its upper block is our real results: the QLoRA Gemma meta-learning model, weighted soft voting, logistic-regression stacking, zero-shot Gemma, and CafeBERT and ViSoBERT retrained on the same files. The lower block lists the published baseline and benchmark performance scores for the UIT-VSMECs from other protocols.

V. Results and Analysis

A. Impact of LLM Augmentation

We trained CafeBERT on three versions of the training set: the original UIT-VSMEC sentences, paraphrases of those sentences, and new sentences for the scarce classes. Table V shows how each version scores on the test set.

TABLE V
IMPACT OF LLM AUGMENTATION ON CAFEBERT (UIT-VSMEC TEST SET)

Training Data	Acc.	W-F1	Fear-F1	Anger-F1	Surprise-F1
Original	0.6344	0.628	0.67	0.38	0.60
Paraphrased	0.6676	0.6682	0.67	0.42	0.61
Generated	**0.6703**	**0.671**	**0.68**	**0.46**	**0.61**

New sentences give the larger gain. Accuracy goes from 63.44% on the original data to 66.76% with paraphrases and 67.03% with generated sentences. Weighted F1 follows the same pattern: 62.8%, 66.82%, and 67.1%. The change is easiest to see on Anger, where CafeBERT was weakest. Anger F1 rises from 38% to 42% with paraphrasing and to 46% with generation. Fear and Surprise were already stronger, so they have less room to move. Fear stays at 67% under paraphrasing and reaches 68% with generation. Surprise moves from 60% to 61% either way.

Table VI then checks those generated sentences against the original posts. Table VII defines the fidelity score used in that comparison.

TABLE VI
QUALITY OF LLM-GENERATED TRAINING SENTENCES

Metric	Original	Generated
Overall fidelity score	—	0.776
Distinct-2	0.593	0.451
Distinct-3	0.930	0.786
Mean sentence length (chars)	56.0	51.0
Train–test leakage (exact / near)	—	0 / 0
Emotion distribution KL div.	—	0.164

Fidelity = mean of stat, distribution, label, and correlation similarity.

No generated sentence matches the test set exactly or nearly, so the gain in Table V is not copied evaluation text. Fidelity is 0.776, the average of four scores between 0 and 1. Statistical similarity compares the average and the spread of sentence length, in characters and in words. Distribution similarity uses the Kolmogorov–Smirnov statistic on those length distributions. Label similarity is $\exp(-D_{\mathrm{KL}})$ on the mix of emotions. $D_{\mathrm{KL}} = 0.164$ bits is the shift from filling every class below 1,500 examples, and that term equals 0.849. Correlation similarity is 0.850. It checks whether length and emotion still move together: r is the Pearson correlation of the two upper triangles, and F is the normalized Frobenius distance between the correlation matrices. Average length stays close, 56 characters against 51, but generated sentences vary much less, and the Kolmogorov–Smirnov tests reject equal length and word-count distributions ($p \approx 0$). Distinct-2 drops from 0.593 to 0.451 and Distinct-3 from 0.930 to 0.786, because we were trying to create more sentences, which used the same Vietnamese vocabulary. The new sentences are a little more uniform than the original posts, and the emotion mix is shifted because the scarce classes were filled in.

TABLE VII
FIDELITY SCORE COMPONENT FORMULA

Component	Definition
Stat. similarity	$1 - \dfrac{\lvert x_{\mathrm{gen}} - x_{\mathrm{orig}} \rvert}{\lvert x_{\mathrm{orig}} \rvert}$
Distribution similarity	$\dfrac{1}{1 + \mathrm{KS}}$
Label similarity	$\exp(-D_{\mathrm{KL}})$
Correlation similarity	$\dfrac{1}{2}(r + (1 - F))$
Fidelity score	mean of the four components

B. Ensemble Performance

The stacked QLoRA model is the strongest system on our split. Table VIII reports 70.33% $\pm$ 1.33 accuracy, 70.39% $\pm$ 1.26 weighted F1, and 67.27% $\pm$ 1.25 macro F1, averaged over seeds 42, 131, and 9998.

Weighted soft voting and logistic-regression stacking reaches 67.91% weighted F1. Logistic regression reaches 67.48%. QLoRA improves on these by 2.48 and 2.91 weighted-F1 percentage points. The macro-F1 gap against logistic regression is wider: 67.27% against 63.27%, a gain of 4.00 percentage points. Both baselines already combine the five base models, so the remaining gain comes from reading the sentence together with their probabilities.

The zero shot prompting score 65.72% weighted F1, 4.67 percentage points behind the fine-tuned model. A single transformer trained on the same files is further behind: CafeBERT reaches 65.10% $\pm$ 0.40 weighted F1 and ViSoBERT 58.05% $\pm$ 0.54, leaving the full ensemble ahead by 5.29 and 12.34 percentage points. Published scores in the lower block of Table VIII come from other preprocessing and evaluation settings, intended to provide additional reference information regarding existing, similar methods..

TABLE VIII
UIT-VSMEC TEST-SET COMPARISON

Method	Acc.	W-F1	M-F1
Same protocol (ours)			
Proposed (stacked + QLoRA Gemma) (best)	**0.7033 ±0.0133**	**0.7039 ±0.0126**	**0.6727 ±0.0125**
Weighted soft voting (ours)	0.6792 ±0.0123	0.6791 ±0.0120	0.6423 ±0.0106
LogReg stacking meta (ours)	0.6768 ±0.0051	0.6748 ±0.0050	0.6327 ±0.0085
Gemma zero-shot (ours)	0.6618 ±0.0029	0.6572 ±0.0029	0.6172 ±0.0010
CafeBERT(ours reproduce result)	0.6522 ±0.0058	0.6510 ±0.0040	0.6207 ±0.0047
ViSoBERT(ours reproduce result)	0.5785 ±0.0060	0.5805 ±0.0054	0.5427 ±0.0073
Published scores (different protocol)			
MLR + preprocessing [10]	0.6436	0.6440	—
CafeBERT [12]	—	0.6612	—
ViSoBERT [13]	0.6810	0.6837	—
XLM-RoBERTa-Large [12]	0.6137	0.6020	—
GRU–CNN–BiLSTM–LSTM [14]	—	—	0.6579

Note. Upper block: same split, preprocessing, and 692-sentence test set. Lower block: previously published scores under other protocols; W-F1: weighted F1; M-F1: macro F1;

VI. CONCLUSION

A. Summary

This paper proposed a two-stage framework for Vietnamese social-media emotion recognition. Stage 1 shows that generated few-shot augmentation improves CafeBERT on the underrepresented emotions more than paraphrasing (Table V). And data augmentation with LLM benefits the performace of models. Stage 2 shows that the QLoRA Gemma meta-learning model, trained on the sentence and the base-model probabilities, outperforms the same-protocol comparators (Table VIII).

B. Limitations & Future Work

1) **Research limitations:** This pipeline trained three different BERT models using three folds to extract probabilities for the final joint data, which requires significant GPU computation. Moreover, fine-tuning the Gemma model also needs the same computational power. All of the experiments were conducted on an NVIDIA A100 80GB GPU.
2) **Cross-dataset and cross-domain generalization:** Future work should evaluate the framework on related Vietnamese NLP tasks (aspect-level sentiment, hate-speech detection, multi-label corpora) to verify that LLM augmentation and stacked ensembling generalize beyond fine-grained emotion recognition.

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